Codio Project File Edit Find View Tools Education Help Configure... Project Index (static) Configure... SANDIPAN-DEY

exerc.py ×

```python
import os
import openai
import secret
from PIL import Image, ImageFilter
from io import BytesIO
import requests
openai.api_key='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'

# WRITE YOUR CODE HERE
from PIL import ImageFilter

# Generate the base image
def generate_base_image(prompt):
    response = openai.Image.create(
        prompt=prompt,
        n=1,
        size="512x512"
    )
    return response['data'][0]['url']

base_image_url = generate_base_image('plane')
img_data = requests.get(base_image_url).content
with open('plane.png', 'wb') as handler:
    handler.write(img_data)

image = Image.open('plane.png')
# Apply an edge enhance filter to the base image
edge_enhanced_image = image.filter(ImageFilter.EDGE_ENHANCE)
```

26% (8:1)

Terminal ×

```
* Management:     https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

* Canonical Livepatch is available for installation.
  - Reduce system reboots and improve kernel security. Activate at:
    https://ubuntu.com/livepatch
*
* Welcome to the Codio Terminal!
```

Guide ×

Collapse

Coding Exercise

Assessment Question

1. Using OpenAI's DALL-E 2, generate an image of a plane in mid-flight. Save the generated image as `plane.png`.

2. Utilize the Python Imaging Library (PIL) to create an edge-enhanced version of the image you generated in the previous step. Save the enhanced image as `enhanced_plane.png`.

In this task, you will use OpenAI's DALL-E 2 to generate an image based on a textual description and then apply an image processing operation using PIL. The task involves two major steps:

1. **Image Generation with DALL-E 2**: Using OpenAI's DALL-E 2, write a function `generate_base_image(prompt)` that takes a text prompt as input and returns a URL for an image generated by DALL-E 2. The image should be 512x512 pixels in size. Download the image and save it as `plane.png`.

2. **Image Processing with PIL**: Next, load the `plane.png` image and apply an edge enhance filter using PIL's `ImageFilter.EDGE_ENHANCE`. Save the resulting edge-enhanced image as `enhanced_plane.png`.

TRY IT!

```
Command was successfully executed.
```

Click here to refresh your plane image

Click here to refresh your enhanced plane image

When ready to submit, please click the button below.

Check It!

LAST RUN on 7/22/2023, 2:33:42 AM

```
Check 1 passed
```

Mark as Completed   Back to dashboard